Smart Healthcare Analytics Web Integration: Technical Framework and Master Prompt Architecture

Clinical Analytics Architecture and Multi-Dimensional Patient Risk Profiling

Modern healthcare data infrastructures require the seamless conversion of complex biometric records into actionable, interactive web intelligence to support early clinical intervention[1]. The *Health360: Comprehensive Healthcare Risk & Disease Analytics* system addresses the delayed identification of severe chronic conditions—such as cardiovascular heart disease, type 2 diabetes mellitus, and ischemic stroke—which frequently results from manual, non-integrated patient record evaluations[1]. Delayed identification exponentially escalates treatment costs, accelerates clinical complications, and degrades overall patient prognosis[1]. By aggregating demographic profiles, physiological vitals, and self-reported behavioral indicators, healthcare systems establish predictive baselines capable of triaging vulnerable patient cohorts before acute diagnostic events occur[1].

The underlying analytical engine relies on a standardized clinical dataset comprising $N = 5,000$ patient records populated across 17 distinct medical and behavioral attributes[1]. To transform continuous physiological data into discrete analytical dimensions, calculated fields are embedded directly within the database schema[1]. The primary continuous-to-categorical transformation segments raw patient age into six life-stage cohorts, providing structured granularity for epidemiological evaluation[1].

Age Range (Years)	Cohort Classification	Physiological Rationale & Analytical Target
18 – 30	Teenager / Young Cohort	Detects early-onset behavioral risk factors, physical inactivity trends, and baseline vital deviations[1].
31 – 40	Young Adult	Captures stress-induced physiological changes, rising body mass indices, and initial hypertensive

		onset[1].
41 – 50	Adult	Establishes critical baseline trends for metabolic syndrome, early diabetes, and pre-hypertensive markers[1].
51 – 60	Middle Age	Evaluates accelerating multi-morbidity interactions, vascular stiffening, and chronic disease progression[1].
61 – 70	Senior	Tracks established chronic disease burdens, treatment efficacy, and secondary complication risks[1].
> 70	Old	Monitors frailty indexes, high-risk stroke probability, and critical multi-organ diagnostic indicators[1].

Cross-sectional analysis across these discretized age cohorts reveals significant epidemiological findings[1]. Average systolic blood pressure displays a tight distribution across all cohorts, consistently residing within the Stage 1 hypertension range ($\approx 133.61 \text{ mmHg}$ to 135.67 mmHg)[1]. While the *Old* cohort exhibits the maximum mean systolic pressure ($\mu_{\text{Old}} = 135.67 \text{ mmHg}$) driven by age-related arterial compliance loss, the narrow variance observed when compared to *Young Adults* ($\mu_{\text{Young Adult}} = 134.67 \text{ mmHg}$) and *Teenagers* ($\mu_{\text{Teenager}} = 134.50 \text{ mmHg}$) confirms that elevated blood pressure is widespread across younger populations[1]. This phenomenon indicates that lifestyle stressors, including physical inactivity, tobacco consumption, and alcohol intake, act as primary drivers of early cardiovascular risk[1].

To present these insights to diverse medical stakeholders, the analytical layout incorporates nine visual components designed to address clinical, analytical, and public health operational roles[1].

Visual Component	Visualization	Metric / Data	Target Clinical

	Format	Domain Represented	Stakeholder & Utility
KPI Metrics Overview	Summary Stat Cards	Total Patients ($5,000$), Diabetes ($1,746$), Heart Disease ($1,499$), Stroke ($1,243$)[1].	**Dr. Michael Carter (Clinical Risk Officer)**: Immediate oversight of macro population disease counts for clinical resource allocation[1].
Blood Pressure Distribution	Single Bar Chart	Mean Blood Pressure across Age Groups (Range: 133.61 —)[1].	**Dr. Michael Carter**: Detection of age-cohort hypertension anomalies and baseline vital shifts[1].
Substance Usage Analysis	Grouped Bar Chart	Alcohol and Tobacco Consumption patterns across 6 Age Groups[1].	**Emily Rodriguez (Healthcare Data Analyst)**: Quantification of behavioral toxin exposures across age cohorts[1].
Physical Activity Level	Donut Chart	Proportion of Active, Moderate, and Sedentary lifestyle behaviors[1].	**Emily Rodriguez**: Assessment of sedentary lifestyle prevalence as a primary modifiable risk factor[1].
Demographic Breakdown	Pie Chart	Patient distribution by Age Group (Teenager: 20.38%, Senior: 16.38%, etc.)[1].	**Dr. Priya Nair (Public Health Researcher)**: Population balancing and demographic vulnerability

			mapping[1].
Metabolic Co-morbidity	Grouped Bar Chart	Heart Disease vs. Diabetes prevalence stratified across 4 BMI Categories[1].	**Dr. Priya Nair**: Evaluation of metabolic interaction effects between obesity and endocrine failure[1].
Cardiac Risk Escalation	Single Bar Chart	Heart Disease Rate by BMI Category (Obese: 39.36%, Overweight: 32.91%)[1].	**Dr. Michael Carter**: Direct quantification of weight-induced cardiac event escalation[1].
Risk Matrix Heatmap	Matrix Heat Map	Heart Disease rate cross-tabulated by BMI Category and Exercise Level[1].	**Emily Rodriguez**: Identification of protective physical activity thresholds across body mass tiers[1].
Diagnostic Symptom Profile	Grouped Bar Chart	Age Group Distribution across Patient-Reported Fatigue Conditions[1].	**Dr. Priya Nair**: Tracking non-specific symptomatic markers as early warning diagnostic flags[1].

Web Embedding Technical Specifications for Tableau Analytics

Integrating dynamic Tableau visualizations into custom web applications requires choosing between legacy iframe embeds, JavaScript API v2 wrapper functions, and native web-component interfaces introduced in the Tableau Embedding API v3[2]. Modern web portals favor the Embedding API v3 standard because it relies on ES6 modules and custom HTML elements, avoiding manual iframe manipulation while allowing complete programmatic control over dynamic filtering, mark selection, and visual event listeners[4].

When embedding content hosted on Tableau Public, standard share URLs must be converted

into embed-ready endpoints by stripping unnecessary query tokens (:sid=, :origin=viz_share_link) and appending structured runtime configuration attributes[2]. The target analytical assets for this system use two primary URLs:

- **Dashboard View Endpoint**: https://public.tableau.com/views/Book1_17930899228350/SmartHealthcareRiskDiseaseDashboard_
- **Data Story Endpoint**: https://public.tableau.com/shared/H49YQ3N5D

Technical Dimension	Standard iFrame Embed	JavaScript API v2	Embedding API v3 (Web Components)
Script Dependency	None (Pure HTML element)[2].	viz_v1.js or tableau-2.min.js[3].	tableau.embedding.3.latest.min.js (ES6 Module)[4].
DOM Integration Tag	<iframe>[2].	with JavaScript instantiation[3].	<tableau-viz> web component tag[4].
Responsive Lifecycle	Fixed pixel or percentage inline styling[2].	Programmatic resize event handlers[3].	Automatic parent container flex-resizing[4].
Filtering Mechanisms	Static URL parameters (?Age+Group=Senior)[7].	Async API calls (applyFilterAsync)[5].	Asynchronous Promises and typed Web API methods[4].
Authentication Support	Session cookies / public view permissions[2].	Ticket-based trusted authentication[3].	Connected Apps (JWT) and modern OAuth 2.0 standards[4].
Implementation Complexity	Low (Static view rendering)[2].	Moderate (Legacy imperative JS)[3].	Low to Moderate (Declarative HTML5 + Modern JS)[4].

User Interface Layout System and Design Specifications

The user interface follows a modern responsive web layout inspired by clean medical

dashboard systems (such as the EasyFolio Bootstrap design template). The layout maintains visual consistency through a central color palette, clear typography hierarchies, and floating capsule navigation elements.



Visual elements are standardized using CSS design tokens to establish clear interface contrast and hierarchy.

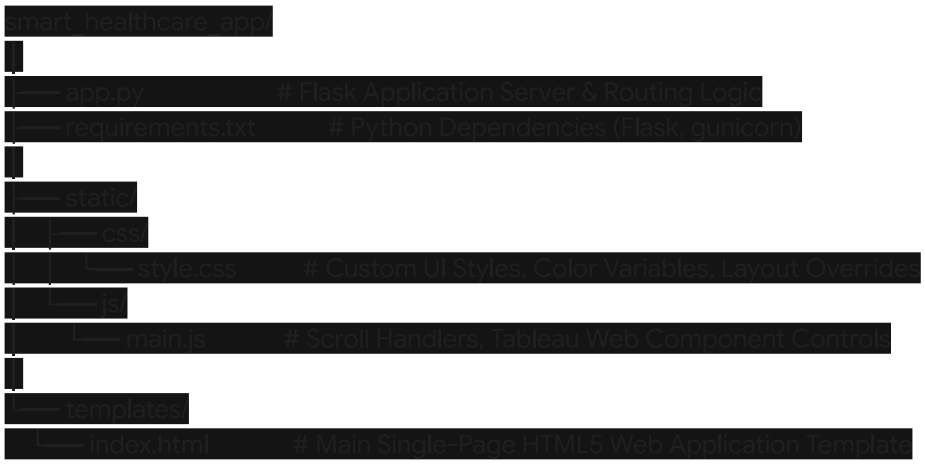
Visual Token Type	CSS Variable / Metric Value	Application Target & UI Context
Primary Accent	#EA580C (Terracotta / Vibrant Orange)	Primary filled call-to-action buttons, active navigation pill markers, interactive link

		highlights.
Secondary Accent	#FFEDD5 (Soft Tinted Amber)	Hover state backgrounds, active tab outline highlights, badge container backgrounds.
Background Pure	#FFFFFF (Solid White)	Core layout cards, floating navigation bar background, interactive modal surfaces.
Background Muted	#F8F9FA / #F3F4F6 (Off-White Soft Grey)	Section background alternating fills, hero container background, FAQ surface fills.
Primary Text	#1E293B (Slate Charcoal)	Major hero headings (<h1>, <h2>), card titles, critical statistical counter numbers.
Muted Text	#475569 (Slate Grey)	Secondary subtitles, body prose paragraphs, metadata grid descriptive labels.
Container Border Radius	50px (Capsule) / 16px (Card)	Rounded floating navigation pills, action buttons, dynamic chart container edges.
Elevation Shadow	0 10px 30px -10px rgba(0,0,0,0.08)	Floating navigation container depth, hover card elevation, visual container framing.

Full-Stack Deployment Strategy and Flask Web Integration

To deliver the integrated web portal, a Python WSGI framework using Flask acts as the core

application engine[1]. Flask serves static UI assets, renders the single-page HTML5 template, handles dynamic client requests, and enables containerized cloud deployment[1].
The application structure follows standard Flask project conventions:



Deploying this architecture to production environments requires specific performance and operational configurations[1]:

- **WSGI Execution Engine**: Execute the Flask service via production-grade WSGI HTTP servers such as Gunicorn (gunicorn --workers 4 --bind 0.0.0.0:5000 app:app), avoiding the single-threaded development server[1].
- **Document Object Model Stability**: Define explicit container aspect ratios (min-height: 850px; width: 100%) within CSS styles to prevent layout shifts while external Tableau JavaScript libraries load[2].
- **Cross-Origin Security & HTTPS**: Enforce HTTPS protocol delivery across all static resources and embedded endpoints to preserve secure cross-origin communication with Tableau Public servers[4].

Master Generation Prompt for Google AI Studio

The prompt below is optimized for direct submission to Google AI Studio (using Gemini 1.5 Pro or Gemini 2.0 Flash). It provides full technical specifications, layout rules, textual descriptions, UI styling directives, and exact Tableau URLs to generate a complete, working codebase (app.py, templates/index.html, static/css/style.css, and static/js/main.js).
You are an expert Full-Stack Engineer and Data Visualization Specialist. Generate a production-ready, highly polished Flask web application for a project titled "Smart Healthcare & Risk Factor Analysis" (also known as Health360).
The web application must faithfully recreate the user interface layout, typography, color

scheme, content structure, and embedded analytics shown in the provided specification and visual mockups.

TECHNICAL STACK & ARCHITECTURE:

1. Backend: Python 3.x using the Flask Web Framework.
2. Frontend: HTML5, CSS3, JavaScript (ES6+), Bootstrap 5.3, and the Tableau Embedding API v3 (tableau-viz custom web component).
3. Project Structure:
   - app.py (Flask server routing)
   - templates/index.html (Single-page web template)
   - static/css/style.css (Custom CSS variables and responsive rules)
   - static/js/main.js (Interactive scroll and script initializations)

EXACT TABLEAU RESOURCE LINKS TO EMBED:

- Tableau Dashboard URL: "https://public.tableau.com/views/Book1_17930899228350/SmartHealthcareRiskDiseaseDashboard_?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link"
- Tableau Story URL: "https://public.tableau.com/shared/H49YQ3N5D?:display_count=n&:origin=viz_share_link"

UI DESIGN SYSTEM & COLOR PALETTE:

- Primary Accent: #EA580C (Vibrant Terracotta / Safety Orange for primary buttons, active pill links, active tab borders).
- Secondary Soft Accent: #FFEDD5 (Light warm tint for button hovers and soft highlights).
- Dark Text / Headers: #1E293B (Deep Slate Charcoal for headings and primary metrics).
- Subtitle / Body Text: #475569 (Muted Slate Grey for body text and descriptive metadata).
- Background Light: #FFFFFF (Pure White for main cards and header background) and #F8F9FA (Off-White for alternating section fills).
- Borders & Radii: 50px pill radius for buttons/nav capsules, 16px radius for layout cards.

PAGE STRUCTURE & CONTENT SPECIFICATIONS:

1. FLOATING NAVIGATION BAR:
   - Floating pill-shaped navigation container centered at the top of the viewport with soft drop-shadow (box-shadow: 0 10px 30px -10px rgba(0,0,0,0.08)).
   - Brand text: "Health Analytics".
   - Links: Home (#home), About (#about), Dashboard (#dashboard), Story (#story), FAQ (#faq).
2. HERO BANNER SECTION (#home):
   - Headline (h1): "Smart Healthcare & Risk Factor Analysis"
   - Sub-headline: "Leveraging data analytics and interactive visualizations to uncover patient risk patterns, lifestyle factors, and healthcare outcomes — empowering smarter clinical decisions."
   - Dual Call-to-Action Buttons:
     - Button 1 (Filled Orange Pill): "View Dashboard" -> Smooth scroll to #dashboard
     - Button 2 (Outline Orange Pill): "Explore Story" -> Smooth scroll to #story

- Key Metrics Summary Counter Bar (3 Columns):
  - Card 1: "10K+" | Subtitle: "Patient Records Analyzed"
  - Card 2: "15+" | Subtitle: "Risk Factors Identified"
  - Card 3: "1" | Subtitle: "Interactive Dashboards"

3. ABOUT SECTION (#about):
   - Header: "About" (Subtitle: "Understanding the data behind patient health risks, lifestyle choices, and clinical outcomes").
   - Subsection Title: "About This Project - Smart Healthcare & Risk Factor Analysis".
   - Project Overview Description:
     "This project explores a comprehensive healthcare dataset to identify key risk factors associated with patient outcomes. Using Tableau for interactive visualization, we analyze how lifestyle habits, demographics, and medical history contribute to health risks such as heart disease, diabetes, and hypertension. The analysis covers patient data across multiple dimensions - including BMI, blood pressure, smoking status, physical activity, and more - to reveal actionable insights that can support healthcare professionals in making data-driven decisions."
   - Project Metadata Grid (2 Rows x 3 Columns of white rounded cards):
     - Item 1: Domain -> Healthcare Analytics
     - Item 2: Tool -> Tableau Public
     - Item 3: Dataset -> Patient Health Records
     - Item 4: Focus Areas -> Risk Factors & Lifestyle
     - Item 5: Visualizations -> Dashboard & Story
     - Item 6: Objective -> Predictive Health Insights
4. INTERACTIVE DASHBOARD SECTION (#dashboard):
   - Header: "Dashboard" (Subtitle: "Interactive dashboard visualizing our data insights and operational metrics").
   - Embedded Element: Include the Tableau Embedding API v3 script (https://public.tableau.com/javascripts/api/tableau.embedding.3.latest.min.js as an ES6 module) and load the exact Dashboard URL using <tableau-viz id="tableauDashboard" src="..." toolbar="bottom" hide-tabs></tableau-viz>.
   - Card container styling: Full width, min-height 850px, subtle shadow, light card border.
5. DATA STORY SECTION (#story):
   - Header: "Story" (Subtitle: "Data story presented in narrative scenes showcasing sequential analytical findings").
   - Embedded Element: Integrate the exact Tableau Story link using <tableau-viz id="tableauStory" src="..." toolbar="bottom"></tableau-viz>.
   - Card container styling: Full width, min-height 850px, subtle shadow, light card border.
6. FREQUENTLY ASKED QUESTIONS SECTION (#faq):
   - Header: "Frequently Asked Questions".
   - Collapsible Accordion containing 3 items:
     1. "What dataset is used in this analytics platform?" -> "The analysis is built on the Smart Healthcare dataset containing 5,000 anonymized patient records structured across 17 clinical and behavioral attributes including blood pressure, cholesterol, BMI, smoking,

and exercise levels."

2. "How are patient age groups categorized?" -> "Patients are grouped using custom calculated fields into 6 cohorts: Teenager (18-30), Young Adult (31-40), Adult (41-50), Middle Age (51-60), Senior (61-70), and Old (>70)."
3. "What clinical roles benefit from this system?" -> "The platform provides tailored views for Clinical Risk Officers (rapid patient triage), Healthcare Data Analysts (lifestyle risk correlation), and Public Health Researchers (population health policy design)."

7. FOOTER SECTION:
   - Clean minimal footer with copyright info: "© 2026 Health360 Smart Healthcare Analytics. All rights reserved."

REQUIRED FILE OUTPUTS:

Provide full, runnable code for each of the following files without omissions or placeholders:

1. app.py
   - Complete Flask application establishing routes for / rendering index.html. Include variables passing Tableau URLs to the jinja template context.
2. templates/index.html
   - Complete single-page HTML5 document incorporating Bootstrap 5.3 CDN, Google Fonts (Plus Jakarta Sans / Inter), FontAwesome icons, Tableau Embedding API v3 script module, and all specified layout sections.
3. static/css/style.css
   - Custom CSS styling implementing the orange/slate color palette, floating capsule navbar styles, hero metric counters, metadata card grid, responsive <tableau-viz> container styles, and media queries for mobile/tablet responsiveness.
4. static/js/main.js
   - JavaScript implementation for smooth anchor scrolling, dynamic floating navbar shadow on scroll, and initializations for Tableau visualization web components.

Works cited

1. Google Keep Document.pdf
2. A Guide to Embedding Tableau # 2: Embed Examples, https://www.flerlagetwins.com/2023/05/embedding-2.html
3. Tableau JavaScript API, https://www.tableau.com/developer/learning/tableau-javascript-api
4. Tableau Embedding API v3, https://help.tableau.com/current/api/embedding_api/en-us/index.html
5. Embed Tableau Dashboard in a JavaScript / React Application, https://wearecommunity.io/communities/india-united-ui-community/articles/1612
6. Tableau Embedding API, https://www.tableau.com/developer/learning/tableau-embedding-api
7. Writing Embed Code for Views - Tableau Help, https://help.tableau.com/current/pro/desktop/en-us/embed_code.htm
8. Embedding API - Tableau, https://www.tableau.com/developer/tools/embedding-api